Employee Attrition & Job Change Prediction System

Project Description Document
Group 5 — DEPI Round 4

1. Project Context & Objectives

Organizations heavily invest time, resources, and capital into hiring and upskilling talent. However, sudden employee departures disrupt ongoing projects and incur substantial replacement costs, often estimated between one to two times an employee's annual salary. The core objective of this project is to develop an end-to-end Machine Learning system that predicts whether an employee is actively looking for a job change. By quantifying attrition risk continuously, HR teams can shift from a reactive stance to a proactive one, intervening with targeted retention strategies (such as compensation adjustments or mentorship) for critical talent before formal resignations occur.

2. Dataset Overview

The system is built upon HR and candidate survey data, detailing professional backgrounds, demographics, and training activity. The dataset is moderately imbalanced, containing an approximate split where 75% of the candidates intend to stay, and 25% are actively seeking a job change.

Key Data Signals Captured:

- • Geographic Data: City development index (a strong indicator of economic mobility).
- • Professional Background: Total experience, relevance of past experience, company size, and type.
- • Educational Background: Education level, major discipline, and university enrollment status.
- • Upskilling Effort: Total training hours completed.

3. Methodological Approach

The project was executed through a structured four-milestone lifecycle designed to transition raw data into a robust, production-ready solution:

1. 1. Data Exploration & Pipeline Architecture: Extensive cleaning protocols were applied to handle missing and malformed data (e.g., treating missing company

information as a predictive signal itself). A reproducible 'ColumnTransformer' pipeline guarantees that all data undergoes exact, isolated standardizations without leakage.

2. 2. Advanced Statistical Analysis & Feature Engineering: Moving beyond raw fields, new signals were synthetically engineered based on statistical testing. Features like 'job mobility score' and the 'experience-to-training ratio' emerged as powerful risk discriminators.
3. 3. Model Development & Selection: A suite of 6 advanced classification algorithms were evaluated. Addressing the class imbalance, LightGBM was ultimately chosen for its superior ability to maximize Recall (identifying the highest percentage of at-risk employees) while maintaining a balanced F1-score. The classification threshold was strategically tuned to 0.5 to prioritize true risk capture over generic precision.
4. 4. Production Deployment & Monitoring: The model is encapsulated in a reproducible artifact structure. It is accessible both via an interactive Streamlit dashboard (designed for HR end-users) and a RESTful FastAPI service (for programmatic integration).

4. Key Predictive Findings

Analytical efforts revealed distinct profiles regarding candidate retention. Most notably, the 'City Development Index' serves as the dominant predictor; candidates residing in cities with lower development indices exhibit a dramatically higher propensity to leave. Additionally, 'Experience Stage' proved highly relevant: Junior professionals (under 5 years) actively enrolled in university courses form the highest-risk demographic profile, whereas Senior experts (10+ years) in established companies show high retention.

5. System Deliverables

The final deliverable is a comprehensive ecosystem that not only scores incoming candidates but tracks its own performance. Key outputs include:

- • The finalized LightGBM predictive model mapped directly into a production pipeline.
- • A Streamlit-based web interface featuring real-time risk predictions with traffic-light alerting.
- • A robust FastAPI endpoint setup, capable of ingesting batch or singular profiles.
- • Comprehensive MLOps integration (using MLflow) to log all prediction instances, parameter configurations, and enable rigorous system monitoring to detect data drift and cue future model retraining.